# MATH3888 Slide 6 Script

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## 1 Main Script

We used PyDSTool to examine how the model changes continuously under the two drug interventions, and for these parameter setups take $[0, 1]$ to be the biologically relevant region.

For TTX, on the upper left, keeping $\alpha_P = 1$, and varying $\alpha_I$, we find a fold bifurcation at around $\mathbf{0.614}$, while a Hopf point lies outside $[0, 1]$.

For ouabain on the upper right, keeping $\alpha_I = 1$ and varying $\alpha_P$, we find **no bifurcations** in the interval $[0, 1]$, and an external Hopf and fold.

For ouabain under complete TTX at the bottom, keeping $\alpha_I = 0$ and varying $\alpha_P$, we actually find a Hopf bifurcation in the interval $[0, 1]$ at around $\mathbf{0.268}$, with the associated periodic-orbit branch, which closes at an external Hopf.

We can then compare these dynamical changes with the experimental data in the following interpretation.

## 2 QnA

**Q1: What do $\alpha_I$ and $\alpha_P$ represent?**

**A1:** They represent the fraction or scaling factor of the TTX-sensitive transient inward conductance and electrogenic pump current remaining as we increase the dose of the drugs respectively. In either case, we have the value 1 for control and 0 for complete blockade.

**Q2: Why is $[0, 1]$ the biologically relevant region?**

**A2:** It is because $\alpha_I$ and $\alpha_P$ are defined as scaling factors for the fraction of the original current remaining. Values on the interval's interior represent partial blockade from the drug; values outside of it do not represent the drug intervention being modelled in a biologically meaningful way.

**Q3: Why did you only include the Hopf envelope for the third bifurcation?**

**A3:** It is because this was the only Hopf bifurcation inside the biologically relevant region. That isn't to say that the external bifurcations play no part in the dynamical changes for TTX or ouabain individually, we just chose not to include every envelope to avoid cluttering the diagrams.

**Q4: Since ouabain had no bifurcations in $[0, 1]$, does it mean that it had no effect?**

**A4:** Not necessarily, it just means that we did not flag any special points that may explain the the qualitative changes in that interval. The external points may be able to explain our transition from a mathematical standpoint, or perhaps a fast-slow analysis can help us do that as well.

**Q5: Does the fold bifurcation for TTX explain its qualitative change to a slow wave oscillation?**

**A5:** Not by itself, and this can suggest that bifurcation analysis on its own is insufficient in explaining our hypotheses. For example, we believe that the slow wave oscillation could be affected by the external Hopf envelopes, or can be explained more with a fast-slow analysis. But as it stands, I would not say the fold on its own can justify the slow wave oscillation; it could be instructive to check the time traces around these points which I believe was done as well for further analysis on TTX.

**Q6: Why did you continue the parameters outside $[0, 1]$, and why did you not use the original parameters to begin with?**

**A6:** We continued them beyond that interval to examine the wider mathematical structure of the equilibrium branch and locate any other potential bifurcations that can explain it. We chose these scaling parameters since it is cleaner for plotting and interpretation to have a unit interval, and it avoids having to deal with signs.

**Q7: Why vary one parameter while fixing the other?**

**A7:** This setup was done for the one-parameter bifurcation studies to isolate the effect of one drug while keeping the other at the untreated control value or at complete blockade, depending on the hypothesis being tested.

**Q8: Did you do a two-parameter bifurcation, and if so, what did you find?**

**A8:** Yes, we did a two-parameter bifurcation on $\alpha_I$ and $\alpha_P$ and did not find any codimension-2 bifurcations within the biologically relevant region, in this case being the unit square. We did find two zero-Hopf points outside the region; that is where the equilibrium point has a zero eigenvalue and a pair of two imaginary eigenvalues.

**Q9: Why are the branches for the TTX fold both unstable?**

**A9:** It the branches are flagged as unstable if at least one eigenvalue has a positive real part, and numerical checks confirm that the largest real part of the eigenvalues of the Jacobian around the fold was positive.